

TrES-1b

R-1403

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_130501 · created 2026-08-01T04:21:02+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456413.83 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.035
Transit depth (Rp/Rs)^2	2.48 +/- 1.12 [%]
Orbital Inclination (inc)	89.41 +/- 1.04
Airmass coefficient 1 (a1)	0.95 +/- 0.015
Airmass coefficient 2 (a2)	0.044 +/- 0.024
Scatter in the residuals of the lightcurve fit is	1.6 %
Best Comparison Star	#1 - [376, 62]
Optimal Aperture	2.71
Optimal Annulus	16.05
Transit Duration (day)	0.0759 +/- 0.0024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.157 $\pm$ 0.035 vs 0.1358 (deviation 0.6 σ)
Duration fit vs published	0.0759 d vs 0.1042 d (deviation 11.59 σ)
Mid-time O-C	-2.6 min
Depth SNR	4.4
Residual scatter	1.6 %

Light curve

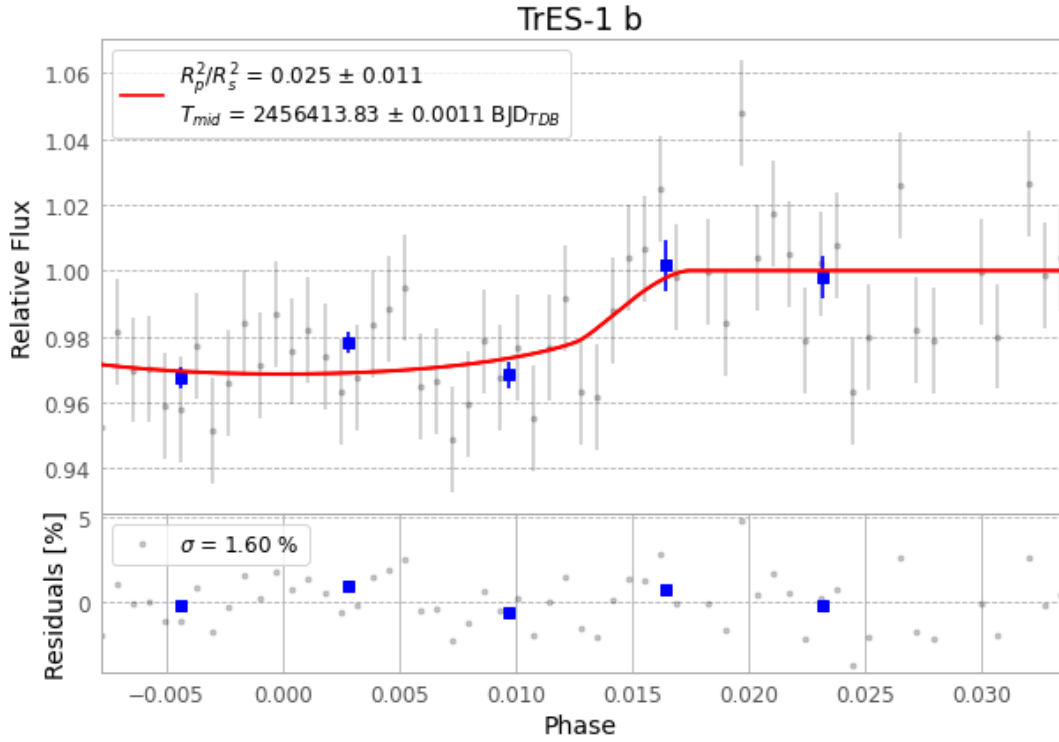


Figure 1 — FinalLightCurve_TrES-1 b_2013-05-01.png (SHA-256 1aee3cf5be55e450...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [289, 195], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5adcd55bbb6e43d0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

58 FITS frames (38 MB), TRES-1130501071818.FITS → TRES-1130501101812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-130501.log (0d6146baca00...), FinalLightCurve_TrES-1 b_2013-05-01.png (1aee3cf5be55...), FinalLightCurve_TrES-1 b_2013-05-01.csv (dba79d42846a...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 04:21Z.